Oh Reum Kwon

ohreumkwon@gmail.com | 408-931-1208 | San Jose, CA | US Citizen

EDUCATION

Georgia Institute of Technology **Aug 2024 - Present**
Master of Science, Computer Science
Relevant Coursework**:** Machine Learning, Computer Vision **Expected: May 2027**

San Jose State University **Aug 2020 - May 2023**
Bachelor of Science, Software Engineering
Relevant Coursework**:** Data Structures & Algorithms, Object Oriented Design, Networks, Operating Systems, DBMS, Security, Enterprise Software

TECHNICAL SKILLS

Quality & Reliability: Statistical Process Control (SPC), Cpk Analysis, Failure Analysis (FA), FMECA, Reliability Testing, Root Cause Analysis
Programming: Python, C++, Java, SQL, MATLAB
ML & Perception: Kalman & Particle Filters, Sensor Fusion (GNSS, IMU, MEMS), Computer Vision
Frameworks & Tools: Claude Code, PyTorch, TensorFlow, OpenCV, NumPy, Docker, Kubernetes, GCP

WORK EXPERIENCE

Apple | Cupertino, CA **Jan 2026 – Aug 2026**
Display Quality Engineer Intern
- Engineered predictive models beyond traditional SPC and Cpk metrics to establish proactive test parameters and evaluate panel quality

SK Hynix Memory Solutions | San Jose, CA **Nov 2025 – Jan 2026**
SSD Reliability Engineer Intern
- Developed automated log parsing scripts in Python, reducing manual data review time by 50%
- Implemented a GPT-2 Large Language Model (LLM) from scratch to generate synthetic AI workloads
- Executed SSD reliability testing, analyzing storage I/O performance and bottlenecks during high-throughput model training

Samsung Semiconductor | San Jose, CA **May 2024 – Aug 2024**
Data Engineer Intern
- Applied ML models to enhance Kalman filter–based algorithms for tunnel detection using GNSS and MEMS data, improving localization accuracy in low-signal environments
- Trained and validated machine learning models in MATLAB to classify tunnel vs. non-tunnel driving conditions with 98% accuracy
- Developed Python automation tools that improved the perception team's workflow efficiency by 30%, by decoding, preprocessing, and visualizing large sensor datasets
- Collaborated with the location team on sensor fusion techniques and data analysis to support advanced positioning and navigation research

PROJECTS

Computer Vision Object Tracking | Python, OpenCV **Dec 2025**
- Building a robust object-tracking system leveraging Kalman and particle filters for multi-object tracking under noise and occlusion
- Exploring applications in autonomous navigation and perception pipelines

AI Trading | Python, scikit-learn **Dec 2024**
- Developed a forecasting system using ensemble learners and Q-learning, integrating technical indicators (SMA, Bollinger Bands, etc.)
- Optimized portfolio strategies, achieving improved returns over baseline models

E-ID | Flutter, Firebase **June 2023**
- Built a multi-factor authentication app with cross-platform support (iOS/Android) for secure ID verification

Pharma Pickup | Java, CSS, SQL **Aug 2022**
- Designed a three-tier architecture for managing prescriptions using Tomcat and MySQL
- Applied concepts in backend data management and system reliability